

The Scots College



Year 12 Mathematics Extension 2

Trial Assessment

August 2006

General Instructions

- All questions are of equal value
- Working time - 3 hours
- Write using blue or black pen
- Board approved calculators may be used
- All necessary working should be shown in every question
- Standard Integrals Table is attached

TOTAL MARKS: 120

WEIGHTING: 40 %

- **Start each question in a new booklet**

QUESTION 1**[15 MARKS]**

(a) For the ellipse $\frac{x^2}{4} + \frac{y^2}{3} = 1$, find

- (i) the eccentricity 1
- (ii) the coordinates of the foci 1
- (iii) the equations of the directrices 1
- (iv) Sketch the ellipse 1

(b) The point $P\left(ct, \frac{c}{t}\right)$ lies on the rectangular hyperbola $xy = c^2$.

- (i) Sketch the hyperbola and mark on it the point P where $t \neq 0$ 1
- (ii) Derive the equation of the tangent at P 2
- (iii) Prove the equation of the normal at P is given by $y = t^2x + \frac{c}{t} - ct^3$ 1
- (iv) The tangent at P meets the line $y = x$ at T. Find the co-ordinates of T. 1
- (v) The normal at P meets the line $y = x$ at N. Find the co-ordinates of N. 1
- (vi) Prove that $OT \times ON = 4c^2$ (O is the Origin) 2

(c) The tangent to the hyperbola at a point P $(a \sec \theta, b \tan \theta)$ on the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ meets the axes at Q and S.

If OQRS is a rectangle, where O is the Origin.

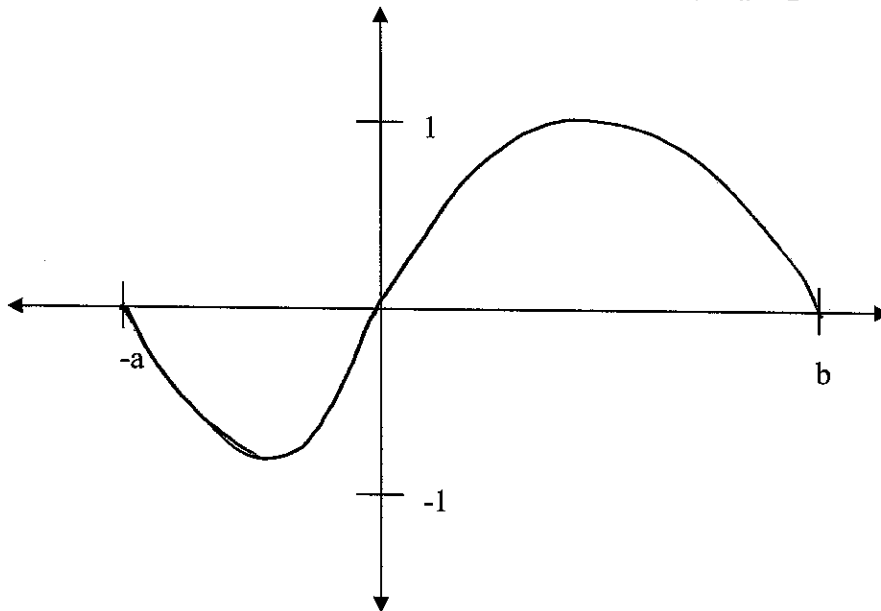
Find the locus of R.

3

QUESTION 2

[15 MARKS]

- (a) The graph of the function $f(x)$ is sketched below.
The domain of the function is $-a \leq x \leq b$ Note $b > a > 1$



On separate number planes sketch the graphs of:

- | | | |
|-------|----------------------|---|
| (i) | $y = f(-x)$ | 1 |
| (ii) | $y = \frac{1}{f(x)}$ | 2 |
| (iii) | $y = e^{f(x)}$ | 2 |
| (iv) | $y = f(x^2)$ | 2 |

- (b) Given $G(x) = \frac{x^2 - 1}{x^2 + 1}$. On separate number planes sketch the graphs of:

- | | | |
|-------|--------------------------------------|---|
| (i) | $y = G(x)$ | 2 |
| (ii) | $[G(x)]^2 = \frac{x^2 - 1}{x^2 + 1}$ | 2 |
| (iv) | $y = \frac{G(x)}{ G(x) }$ | 2 |
| (iii) | $y = \frac{ x+1 (x-1)}{x^2 + 1}$ | 2 |

QUESTION 3**[15 MARKS]**

(a) Given $z = 1 - i$, find:

(i) $\operatorname{Im}\left(\frac{1}{z}\right)$ 2

(ii) z^8 in the form $x + yi$ 2

(iii) two values of w such that $w^2 = 3\bar{z} + i$ 3

(b)

(i) Find the five fifth roots of unity 2

(ii) If $w = \operatorname{cis} \frac{2\pi}{5}$, show that $1 + w + w^2 + w^3 + w^4 = 0$ 1

(iii) Show that $z_1 = w + w^4$ and $z_2 = w^2 + w^3$ are roots of the equation $z^2 + z - 1 = 0$ 2

(c) In the Argand diagram the points A, B, C and D represent the complex numbers α, β, λ and δ respectively.

(i) Describe the point which represent $\frac{1}{2}(\alpha + \lambda)$ 1

(ii) Deduce that if $\alpha + \lambda = \beta + \delta$ then ABCD is a parallelogram. 2

QUESTION 4**[15 MARKS]****(a)** Find the exact value of:

(i) $\int_0^{\pi} \sin^3 x \, dx$ 2

(ii) $\int_1^2 \frac{(\log_e x)^2}{x} dx$ 2

(iii) $\int_0^{\frac{1}{2}} \cos^{-1} x \, dx$ 2

(b) Use the substitution $u = 2 + \cos \theta$ to show that

$$\int_0^{\frac{\pi}{2}} \frac{\sin 2\theta}{2 + \cos \theta} d\theta = 2 + 4 \log_e \left(\frac{2}{3} \right)$$
 3

(c) By using $\tan\left(\frac{x}{2}\right) = t$, integrate

$$\int \sec x dx$$
 2

(d)

(i) Explain why $\int_0^a f(x) dx = \int_0^a f(a-x) dx$ 2

(ii) Hence show that $\int_0^{\frac{\pi}{2}} \frac{\sin^m x}{\sin^m x + \cos^m x} dx = \frac{\pi}{4}$ 2

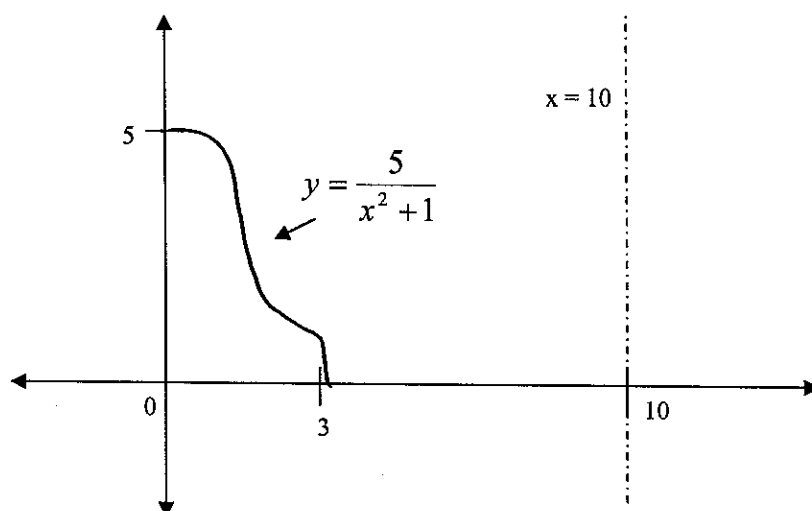
QUESTION 5

[15 MARKS]

- (a) A polynomial function is $P(x) = x^5 + x^4 + 13x^3 + 13x^2 - 48x - 48$.
Factorise $P(x)$ over the field of:
- (i) real numbers 2
 - (ii) imaginary numbers 1
- (b) Solve $2x^4 + 9x^3 + 6x^2 - 20x - 24 = 0$ if it has a root of multiplicity of 3. 3
- (c) Let α, β and λ be the roots of a cubic equation $x^3 + px^2 + q = 0$, where p and q are real.
The equation $x^3 + ax^2 + bx + c = 0$ has the roots α^2, β^2 and λ^2 .
Find a, b and c in terms of p and q 3
- (d) A monic cubic polynomial, when divided by $x^2 + 4$ it leaves a remainder of $x + 8$.
When it is divided by x it leaves a remainder of -4 .
Find the polynomial in expanded form 2
- (e) By using De Moivre's theorem, show that the expansion of
 $\cos 5\theta = 16\cos^5 \theta - 20\cos^3 \theta + 5\cos \theta$ 2
- Hence solve the equation $16x^4 - 20x^2 + 5 = 0$ 2

QUESTION 6
(a)

[15 MARKS]



A circular flange is formed by rotating the region bounded by the curve $y = \frac{5}{x^2 + 1}$, the x axis and the lines $x = 0$ and $x = 3$, about the line $x = 10$.
(All measurements are in cm)

- (i) Use the **method of cylindrical shells** to show the volume generated $V \text{ cm}^3$ of the flange is given by $V = \int_0^3 \frac{(100 - 10x)\pi}{x^2 + 1} dx$. 2

- (ii) Hence find the volume of the flange correct to the nearest cm^3 2

- (b) Use the **method of slices** to find the volume generated when the area bounded by $y = x^2 - 3x^4$ and the x-axis is rotated about the y-axis. Begin with a suitable sketch.

- (i) Evaluate the volume. 4

- (ii) Give two reasons why using the “**method of cylindrical shells**” in this question would have been easier. 1

- (iii) Use the **method of cylindrical shells** to evaluate the volume 2

- (c) Show the area of an isosceles right angled triangle with hypotenuse h is given by $\frac{h^2}{4}$ 1

The base of a solid is the region enclosed by the curve $y = 9 - x^2$ and the x-axis. Each cross-section perpendicular to the y-axis is an isosceles right angled triangle with hypotenuse lying in the base.
Use integration to find the volume of the solid. 3

QUESTION 7

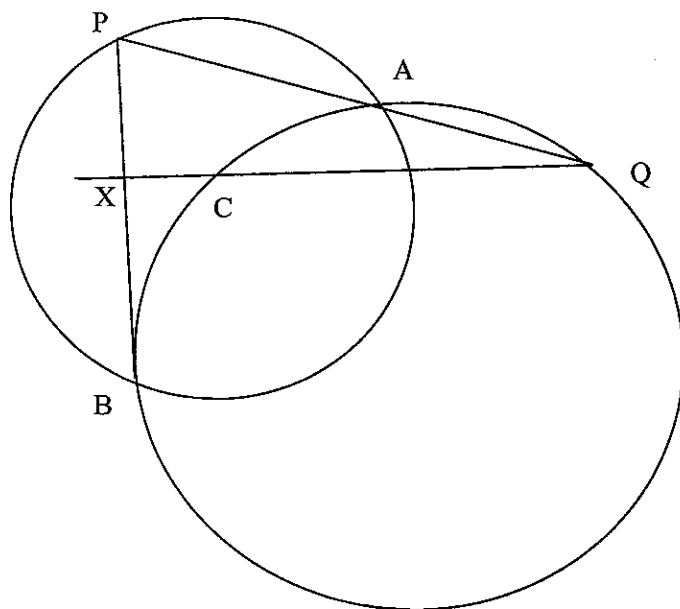
[15 MARKS]

(a) $I_n = \int_0^{\frac{\pi}{2}} \cos^n x dx \quad n \geq 0$

(i) Show that $I_n = \frac{n-1}{n} I_{n-2}$ 2

(ii) Hence evaluate $\int_0^{\frac{\pi}{2}} \cos^4 x dx$ 2

(b)



The two circles intersect at A and B. the larger circle passes through the centre, C, of the smaller circle. P and Q are points on the circle such that PQ passes through A. QC is produced to meet PB at X

Make a neat copy of the diagram on your answer sheet. Let $\angle QAB = \theta$

(i) Show that $\angle BCX = 180^\circ - \theta$ 2

(ii) Prove that $\angle PXC = 90^\circ$ 3

(c)

(i) If a and b are positive real numbers prove $\frac{a+b}{2} \geq \sqrt{ab}$ 1

(ii) Hence if a, b and c are positive real numbers prove $(a+c)(b+c)(b+a) \geq 8abc$ 1

Hence if a, b and c are sides of a triangle, assuming that $0 < c \leq b \leq a$, prove that;

(iii) $(a+b-c)(b+c-a)(c+a-b) \leq abc$ 2

(iv) $a^2(b+c-a) + b^2(c+a-b) + c^2(a+b-c) \leq 3abc$ 2

QUESTION 8**[15 MARKS]**

A sequence U_n has the general term $\frac{1}{(n+1)(n+2)}$ for all integers $n \geq 1$

- (i) Prove by Induction that $\sum_{k=1}^n \frac{1}{(k+1)(k+2)} = \frac{n}{2(n+2)}$ **3**
- (ii) Evaluate $\lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{1}{(k+1)(k+2)}$. Let this be called S **2**
- (iii) Evaluate $\int_1^{\infty} \frac{dx}{(x+1)(x+2)}$. Let this be called I **3**
- (iv) Sketch the graph of the function $y = \frac{1}{(x+1)(x+2)}$ for $x \geq 0$ **2**
- (v) Using the Trapeziodal Rule, with equal strips of unit width, find and approximate area under the the function $y = \frac{1}{(x+1)(x+2)}$ for $x \geq 0$. Let this be called T **3**
- (vi) Using graphic means, explain the relationship between S, I and T, in terms of their size. **2**

END OF PAPER

Standard Integrals

$$\int x^n dx = \frac{1}{n+1} x^{n+1} + C, \quad n \neq -1; x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x + C, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax} + C, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax + C, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax + C, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax + C, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax + C, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a} + C, \quad a \neq 0$$

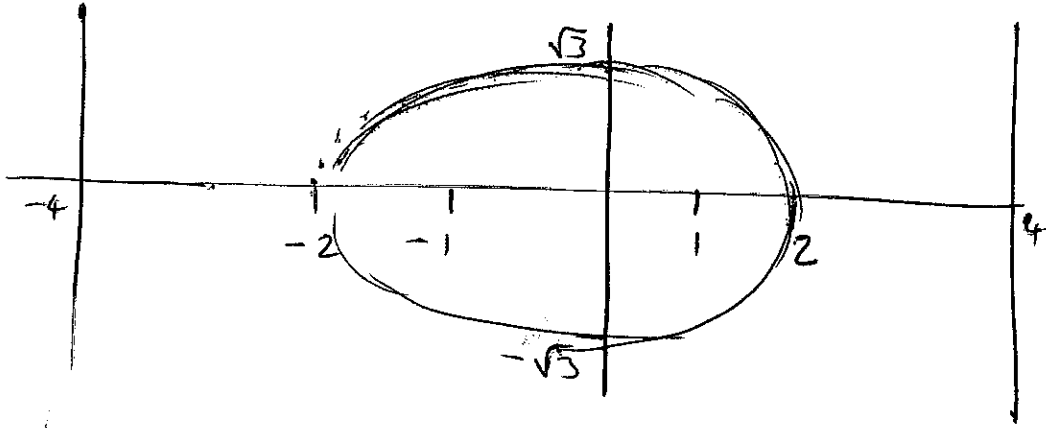
$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a} + C, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right) + C, \quad x > a > 0$$

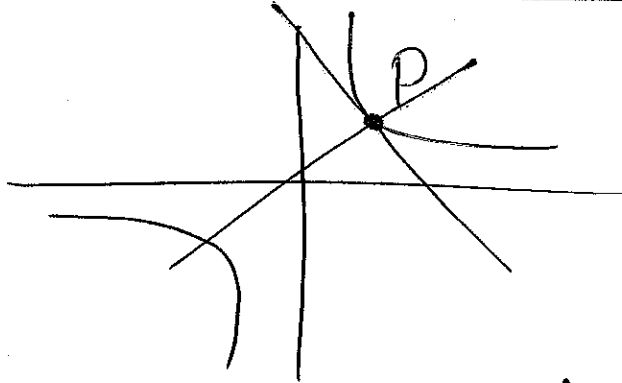
$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right) + C$$

NOTE : $\ln x \equiv \log_e x, \quad x > 0$

QVI a) $e = \frac{1}{2}$, Foci $(\pm 1, 0)$, Directrix $x = \pm 4$



b)



$$y = \frac{c^2}{x}$$

$$y' = -\frac{c^2}{x^2}$$

$$= \frac{-c^2}{(ct)^2} = -\frac{1}{t^2}$$

$$y - \frac{c}{t} = -\frac{1}{t^2}(x - ct)$$

$$x + yt^2 - 2ct = 0$$

Normal

$$y' = t^2$$

$$y - \frac{c}{t} = t^2(x - ct)$$

$$y = t^2x + \frac{c}{t} - ct^3$$

$$N \left(\frac{c(1+t^2)}{t}, \frac{c(1+t^2)}{t} \right)$$

$$ON = \sqrt{2} \frac{c(1+t^2)}{t}$$

$$OT = \sqrt{2} \frac{2ct}{1+t^2}$$

$$ON \cdot OT = 4c^2 \quad P \in D$$

$$\left(\frac{2ct}{(1+t^2)}, \frac{2ct}{(1+t^2)} \right)$$

c)

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

$$\frac{2x}{a^2} - \frac{2y}{b^2} \frac{dy}{dx} = 0$$

$$m = \frac{b \sec \theta}{a \tan \theta}$$

Tangent $\frac{\sec \theta x}{a} - \frac{\tan \theta y}{b} = 1$

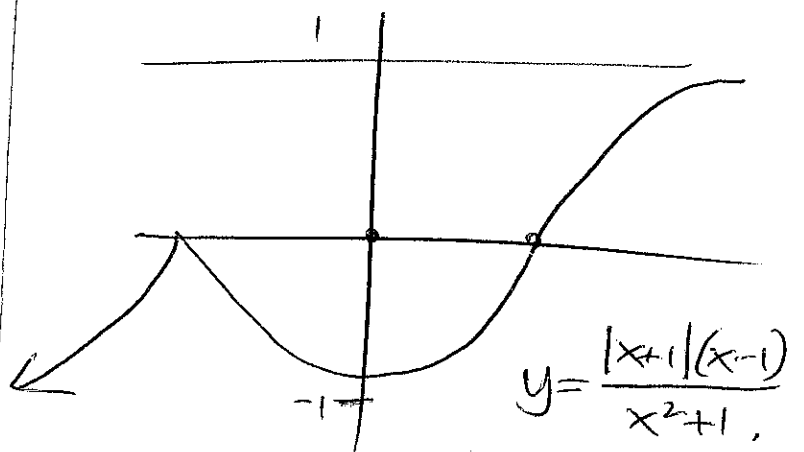
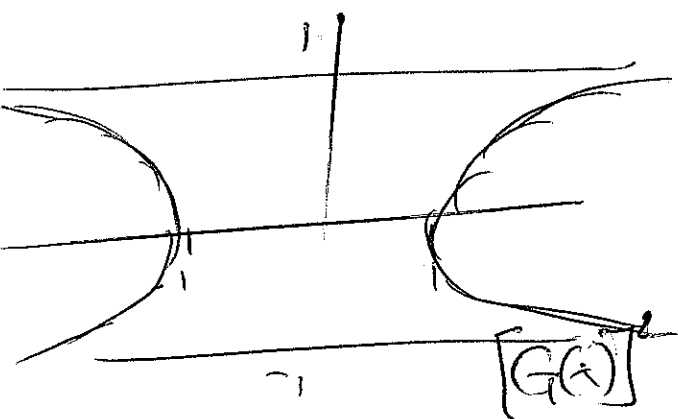
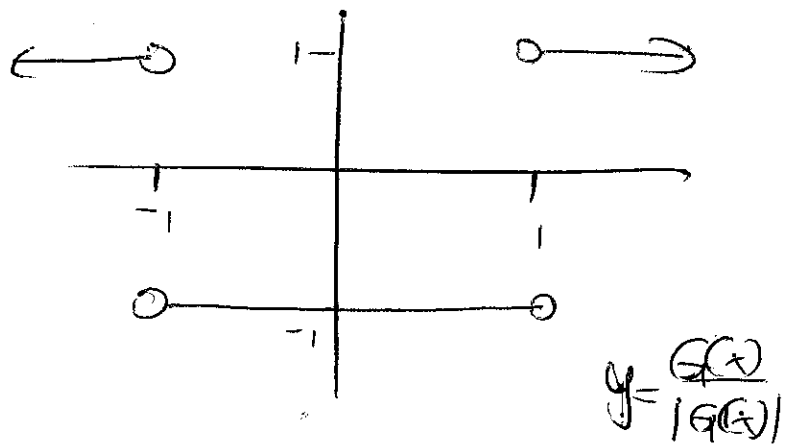
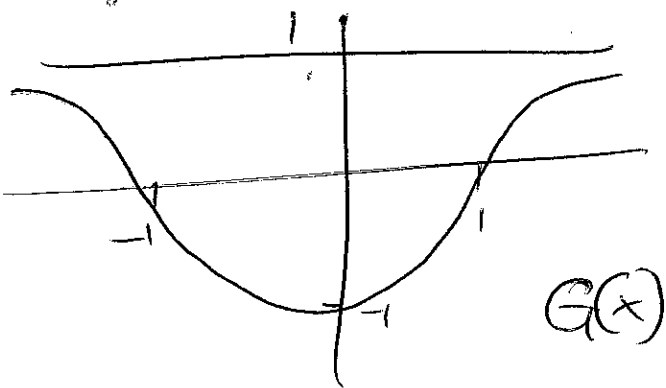
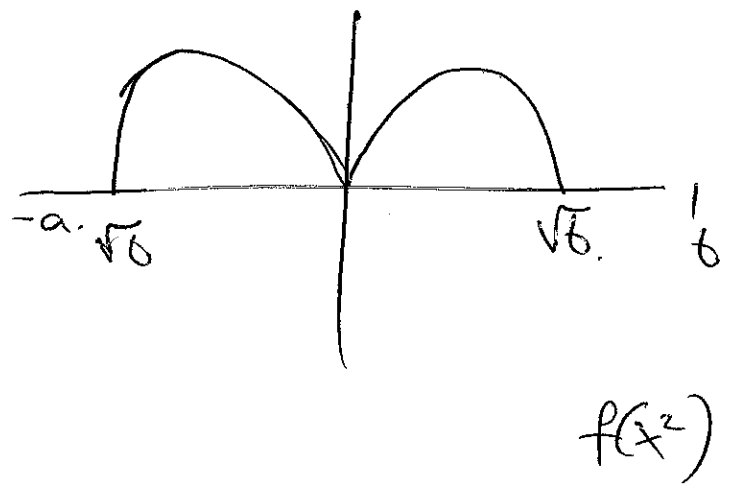
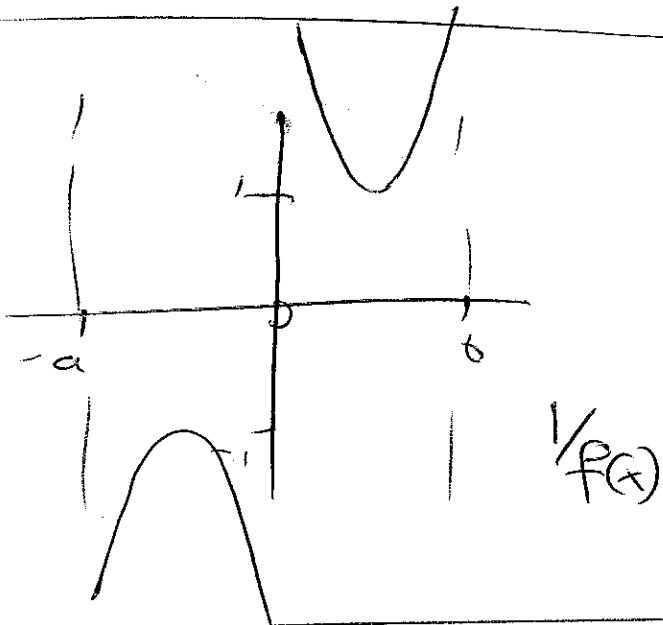
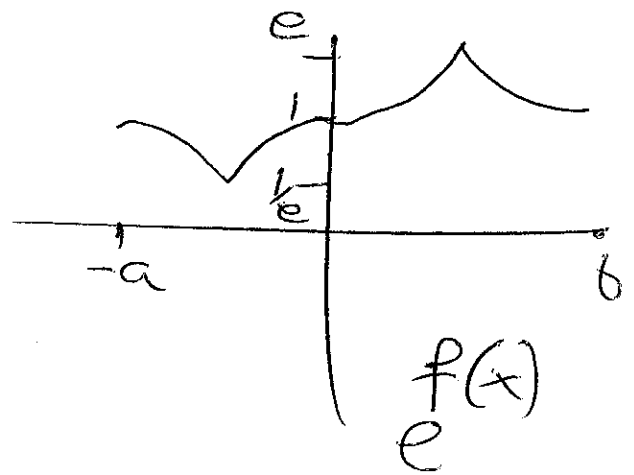
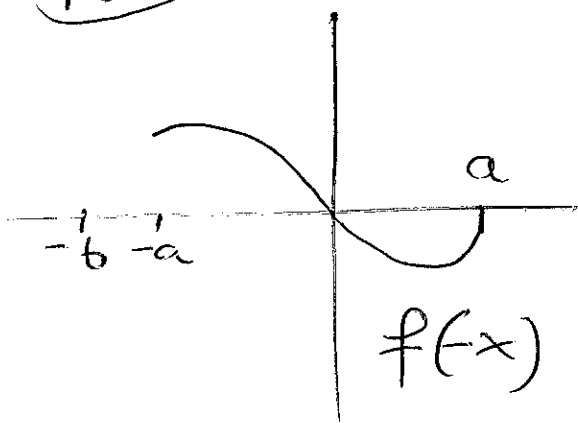
$$x=0 \quad y = -\frac{b}{\tan \theta}$$

$$y=0 \quad x = a \sec \theta$$

$$[\sec^2 \theta - \tan^2 \theta = 1]$$

Locus $\frac{a^2}{x^2} - \frac{b^2}{y^2} = 1$

Q02



Q03

a) i)

$$Z = 1 - i$$

$$\frac{1}{Z} = \frac{1}{1-i}$$

$$= \frac{1}{1-i} \times \frac{1+i}{1+i}$$

$$= \frac{1}{2} + \frac{1}{2}i$$

$$\therefore \text{Im}\left(\frac{1}{Z}\right) = \frac{1}{2}$$

ii)

$$Z = \sqrt{2} \text{cis} -\frac{\pi}{4}$$

$$Z^8 = \sqrt{2}^8 \text{cis} \left(-\frac{\pi}{4} \cdot 8\right)$$

$$= 16 \text{cis} (2\pi)$$

$$= 16 + 0i$$

iii)

$$W^2 = 3(1+i) + i$$

$$= 3 + 4i$$

$$\sqrt{3+4i} = x + yi$$

$$3+4i = (x+yi)^2 = x^2 - y^2 + 2xyi$$

$$x^2 - y^2 = 3 \quad 2xy = 4$$

$$x^4 + 3x^2 - 4 = 0$$

$$(x^2 + 4)(x^2 - 1) = 0$$

Not Real

$$y = \pm 1 \Rightarrow x = \pm 2$$

$$W = 2+i, -2-i$$

b) CIS O

$$\therefore W = \text{cis} 0$$

$$W_1 = \text{cis} \frac{2\pi}{5}$$

$$W_2 = \text{cis} \frac{4\pi}{5}$$

$$W_3 = \text{cis} \frac{6\pi}{5}$$

$$W_4 = \text{cis} \frac{8\pi}{5}$$

iii)

$$\text{Sum of Roots} = -1$$

$$\text{Product of Roots} = 1$$

$$(W+W^2+W^3+W^4) = -1$$

$$(W+W^4)(W^2+W^3)$$

$$W^3+W^4+W^6+W^2 = -1$$

W

$$ii) W^5 = 1$$

$$W^5 - 1 = 0$$

$$(W-1)(W^4+W^3+W^2+W+1) = 0$$

$$= 0$$

c)

i) Mid Point of Line



$$ii) \frac{\alpha + \beta}{2} = \frac{\beta + \alpha}{2}$$

Mid Points at same point

Q04 $\int_0^{\pi} \sin^3 x dx$

$$\int_0^{\pi} \sin x \sin^2 x dx$$

$$\int_0^{\pi} \sin x (1 - \cos^2 x) dx$$

$$\int \sin x - \sin x \cos^2 x dx$$

$$\left[-\cos x - \frac{1}{3} \cos^3 x \right]_0^{\pi}$$

$$-\frac{4}{3}$$

$$\int_1^2 \frac{(\log x)^2}{x} dx$$

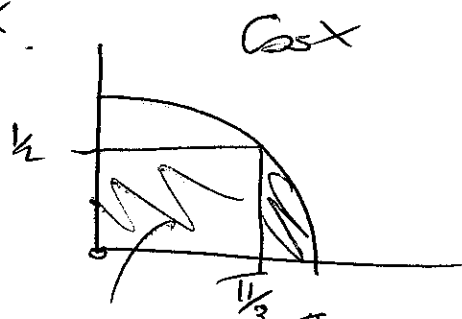
$$\left(\frac{(\log x)^3}{3} \right)_1^2$$

$$\frac{1}{3} (\ln 2)^3$$

$\int_0^{\frac{1}{2}} \cos^{-1} x dx$ By Parts
or $\cos x$.

$$V = \cos^{-1} x \quad du = 1$$

$$dv = \frac{-1}{\sqrt{1-x^2}} \quad U = x$$



$$\frac{\pi}{6} + \int_{\frac{\pi}{3}}^{\frac{\pi}{2}} \cos x dx$$

$$\frac{\pi}{6} + [\sin x]_{\frac{\pi}{3}}^{\frac{\pi}{2}}$$

$$\frac{\pi}{6} + 1 - \frac{\sqrt{3}}{2}$$

$$\int_0^{\frac{\pi}{2}} \frac{2 \sin 2\theta}{2 + \cos \theta} d\theta$$

$$\int_0^{\frac{\pi}{2}} \frac{2 \sin \theta \cos \theta}{2 + \cos \theta} d\theta$$

$$\int_3^2 \frac{2 - U}{U} dU$$

$$\int_3^2 \left(\frac{2}{U} - 1 \right) dU$$

$$U = 2 + \cos \theta$$

$$dU = -\sin \theta d\theta$$

$$d\theta =$$

$$\left[2 \ln U - U \right]_3^2$$

QED

4c)

$$\int \sec x dx = \int \frac{1}{\cos x} dx$$

$$\cos x = \frac{1+t^2}{1-t^2}$$

$$dx = \frac{1}{2} \sec^2 \frac{x}{2} dx$$

$$= \frac{1}{2} (1+t^2) dt$$

$$dx = \frac{2dt}{1+t^2}$$

$$\int \frac{1+t^2}{1-t^2} \cdot \frac{2dt}{1+t^2}$$

Partic.

$$\int \frac{1}{1-t} + \frac{1}{1+t} dt$$

$$\log \frac{(1+t)}{(1-t)} + C \equiv \log \frac{(1+\tan \frac{x}{2})}{(1-\tan \frac{x}{2})} + C$$

$$\int_0^a f(x) dx$$

$$\text{let } u = a-x$$

$$\therefore \text{ Limits } a \Rightarrow 0$$

$$0 \Rightarrow a$$

$$\frac{du}{dx} = -1$$

$$= \int_a^0 f(u) du$$

$$= a \int_0^a f(u) du$$

or by
diagram.

$$I = \int_0^{\pi/2} \frac{\sin^m x}{\sin^m x + \cos^m x} dx$$

$$= \int_0^{\pi/2} \frac{\sin^m (\frac{\pi}{2} - x)}{\sin^m (\frac{\pi}{2} - x) + \cos^m (\frac{\pi}{2} - x)} dx$$

$$I = \int_0^{\pi/2} \frac{\cos^m(x)}{\cos^m(x) + \sin^m(x)} dx$$

$$2I = \int_0^{\pi/2} \frac{\sin^m x}{\sin^m x + \cos^m x} + \frac{\cos^m x}{\cos^m x + \sin^m x} dx$$

$$2I = \int_0^{\pi/2} 1 dx$$

$$= [x]_0^{\pi/2}$$

$$2I = \frac{\pi}{2} \Rightarrow I = \frac{\pi}{4}$$

5

$$a) P(-1) = 0$$

$x+1$ is a factor

$$x+1 \overline{) x^4 + 13x^2 - 48}$$

$$(x+1)(x^4 + 13x^2 - 48)$$

$$(x+1)(x^2 + 16)(x^2 - 3)$$

I R

$$(x+1)(x-\sqrt{3})(x+\sqrt{3})(x^2+16)$$

$$(x+1)(x-\sqrt{3})(x+\sqrt{3})(x-4i)(x+4i)$$

$$b) 2x^4 + 9x^3 + 6x^2 - 20x - 24 = 0$$

$$\text{mult } 3 \quad \therefore f(x)=0 \quad f'(x)=0 \quad f''(x)=0$$

$$24x^2 + 54x + 12 = 0$$

$$(4x+1)(x+2)$$

$$x = -1/4 \quad x = -2$$

One is triple root, one is turn point only

$$P(-2) = 0$$

$$(x+2)(x+2)(x+2)(ax+b)$$

$$\text{By inspect. } a=2, b=-3$$

$$\therefore \text{Solve } x = -2, -3/2$$

$$c) P(x) = x^3 + px^2 + q = 0 \quad \alpha, \beta, \lambda.$$

$$\text{For } \alpha^2, \beta^2, \lambda^2 \quad P(x^{\frac{1}{2}}) = (x^{\frac{1}{2}})^3 + p(x^{\frac{1}{2}})^2 + q$$

$$x^{\frac{3}{2}} + px + q = 0$$

$$x^{\frac{3}{2}} = -px - q$$

$$(x^{\frac{3}{2}})^2 = (-px - q)^2$$

$$x^3 = p^2x^2 + 2pqx + q^2$$

$$x^3 - p^2x^2 - 2pqx - q^2$$

$$0x^3 + ax^2 + bx + c$$

$$a = -p^2 \quad b = -2pq \quad c = -q^2$$

$$d) P(x) = 1x^3 + ax^2 + bx + c$$

$$P(x) = x^2 + 4Q(x) + x + 8$$

$$\text{By deduction } Q(x) = \textcircled{5}(x+a)$$

$$(x^2 + 4)(x+a) + x + 8$$

$$x^3 + ax^2 + \textcircled{4x+x} + 4a + 8$$

$$\therefore b = 5 \quad \text{and} \quad 4a + 8 = c$$

$$P(x) = xR(x) - 4$$

$$\text{By deduc} R(x) = x^2 + ax + b$$

$$x(x^2 + ax + b) - 4$$

$$x^3 + ax^2 + bx - 4$$

$$\therefore c = -4$$

$$4a + 8 = c \Rightarrow a = -3$$

$$a = -3, \quad b = 5, \quad c = -4$$

$$e)(CIS\theta)^5 = 1^5(\cos 5\theta + i\sin 5\theta)$$

By De Moivre's

$$\begin{array}{ccccccc} & & 1 & & 1 & & \\ & 1 & & 3 & & 3 & & 1 \\ 1 & & 3 & & 3 & & 1 \\ 1 & 4 & 6 & 4 & 1 \\ 1 & 5 & 10 & 10 & 5 & 1 \end{array}$$

$$\therefore (CIS\theta)^5 = \underbrace{C^5}_{\text{Real}} + 5C^4iS + \underbrace{10C^3i^2S^2}_{\text{Real}} + \underbrace{10C^2i^3S^3}_{\text{Real}} + \underbrace{5Ci^4S^4}_{\text{Real}} + \underbrace{i^5S^5}_{\text{Real}}$$

$$\begin{aligned} \cos(5\theta) &= \cos^5\theta + 10C^3S^2 + 5CS^4 \\ &= C^5 - 10C^3(1-C^2) + 5C(1-C^2)^2 \\ &= C^5 - 10C^3 + 10C^5 + 5C - 10C^3 + 5C^5 \\ &= 16C^5 - 20C^3 + 5C \end{aligned}$$

$$\cos 5\theta = C(16C^4 - 20C^2 + 5)$$

When does this equal 0, $\Rightarrow \cos 5\theta = 0$

$$\therefore 5\theta = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2}, \frac{7\pi}{2}, \frac{9\pi}{2}$$

$$\theta = \frac{\pi}{10}, \frac{3\pi}{10}, \frac{5\pi}{10}, \frac{7\pi}{10}, \frac{9\pi}{10} \quad [\text{Repeats}]$$

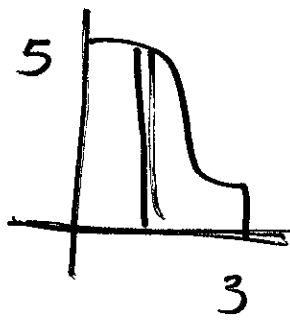
$$\begin{aligned} 5 \text{ Solution } &\Rightarrow 1 \rightarrow C \\ &4 \rightarrow 16C^4 - 20C^2 + 5 \end{aligned}$$

$$\cos \frac{5\pi}{10} = \cos \frac{\pi}{2} = 0 \quad \text{This is the } C \text{ solution}$$

(As a check $16x^4 - 20x^2 + 5 \neq 0$ when $x = \cos \frac{\pi}{10} = 0$.)

$$\text{Solut } x = \cos \frac{\pi}{10}, \cos \frac{3\pi}{10}, \cos \frac{7\pi}{10}, \cos \frac{9\pi}{10}$$

6 a)



$$\text{Radius} = (10 - x)$$

$$\text{Height} = y$$

$$V = 2\pi \int_0^3 (10 - x)y \, dx$$

$$V = 2\pi \int_0^3 (10 - x) \frac{5}{x^2 + 1} \, dx$$

$$V = \int_0^3 \frac{(100 - 10x)\pi}{x^2 + 1} \, dx$$

$$V = 10\pi \int_0^3 \frac{10 - x}{x^2 + 1} \, dx$$

$$= 10\pi \int_0^3 \left(\frac{10}{x^2 + 1} - \frac{x}{x^2 + 1} \right) \, dx$$

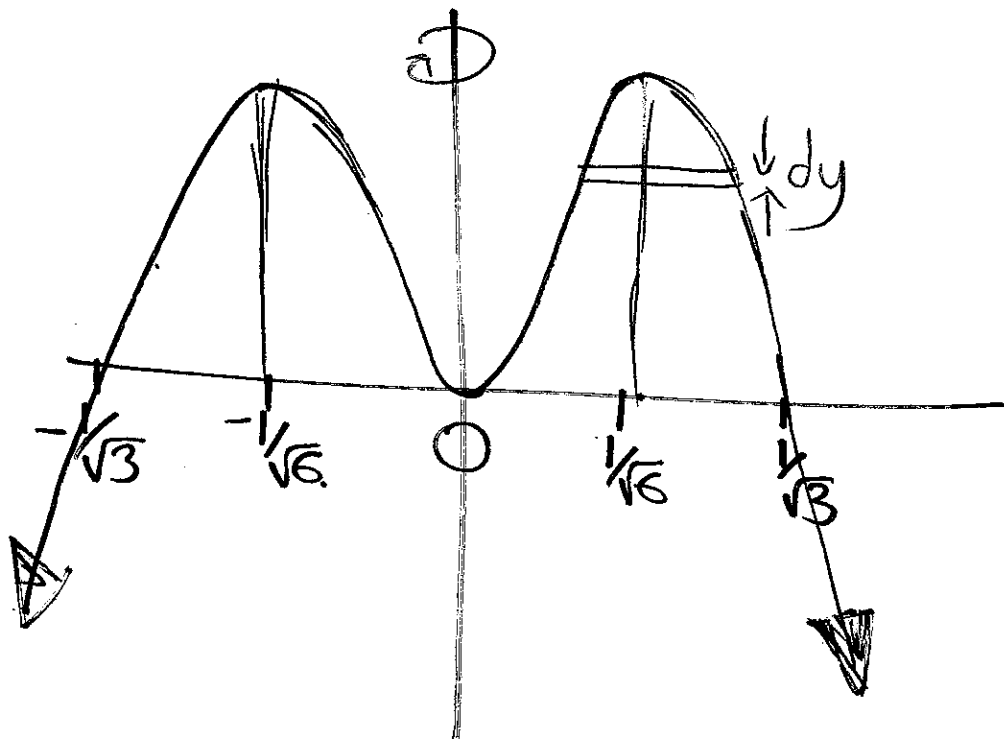
$$= 10\pi \left[10 \tan^{-1} x - \frac{1}{2} \ln(x^2 + 1) \right]_0^3$$

$$= 10\pi \left[10 \tan^{-1} 3 - \frac{1}{2} \ln 10 \right]$$

2

nearest cm.

b)



$$x^2(1 - 3x^2) = 0$$

$$x^2 = 0 \quad 3x^2 = 1$$

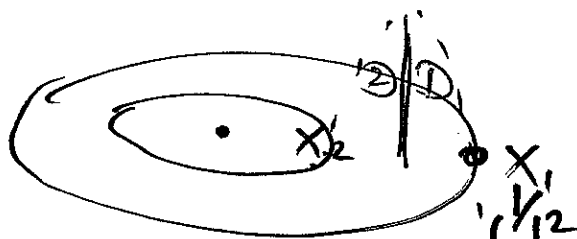
$$x = 0 \quad x = \pm \frac{1}{\sqrt{3}}$$

$$y = x^2 - 3x^4$$

$$\frac{dy}{dx} = 2x - 12x^3$$

$$0 = 2x(1 - 6x^2)$$

$$x = 0 \quad x = \pm \frac{1}{\sqrt{6}}$$



$$V = \pi \int_0^{y_{12}} [x_1^2 - x_2^2] dy$$

Split y_{12} into two parts: ① and ②

$$\pi \int_0^{y_{12}} x^2 - \left(\frac{y}{\sqrt{6}}\right)^2 dy + \pi \int_0^{y_{12}} \left(\frac{y}{\sqrt{6}}\right)^2 - x^2 dy$$

Change to dx

$$\frac{dy}{dx} = 2x - 12x^3$$

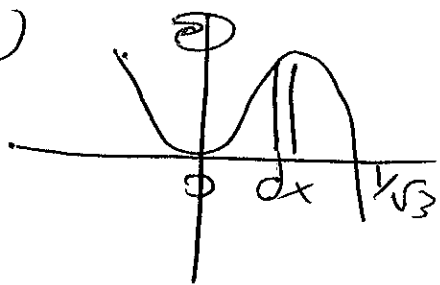
$$= \pi \int_{\frac{1}{\sqrt{3}}}^{\frac{1}{\sqrt{6}}} (x^2 - \frac{1}{6})(2x - 12x^3) dx + \pi \int_0^{\frac{1}{\sqrt{6}}} \left(\frac{y}{6}\right) - x^2 (2x - 12x^3) dx$$

$$= \pi \int_{\frac{1}{\sqrt{3}}}^{\frac{1}{\sqrt{6}}} \left(4x^3 - 12x^5 - \frac{x}{3}\right) dx + \pi \int_0^{\frac{1}{\sqrt{6}}} \left(\frac{x}{3} - 4x^3 + 12x^5\right) dx$$

$$= \pi \frac{1}{54} 0^3$$

- i) Difficult to connect x_1 and x_2
 Need to change to dx and hence limits
 Much more complicated algebra

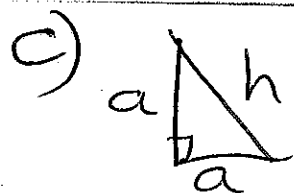
b) iii)



height = y

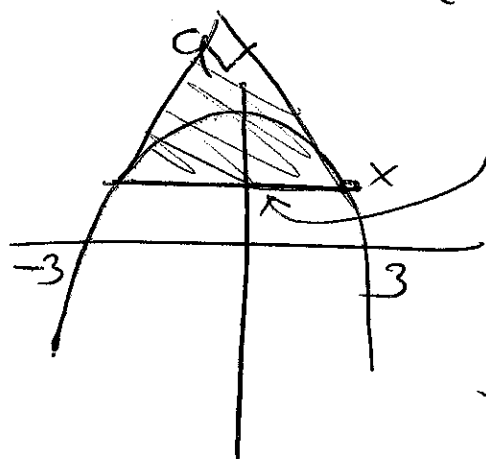
radius = x

$$\begin{aligned}
 V &= 2\pi \int_0^{\sqrt{3}} xy \, dx \\
 &= 2\pi \int_0^{\sqrt{3}} x(x^2 - 3x + 4) \, dx \\
 &= 2\pi \int_0^{\sqrt{3}} (x^3 - 3x^2 + 4x) \, dx \\
 &= 2\pi \left[\frac{x^4}{4} - \frac{3x^3}{3} + \frac{4x^2}{2} \right]_0^{\sqrt{3}} \\
 &= \frac{\pi}{54} V^3
 \end{aligned}$$



$$\begin{aligned}
 a^2 + a^2 &= h^2 \\
 \therefore 2a^2 &= h^2 \\
 a^2 &= \frac{h^2}{2} \\
 a &= \frac{h}{\sqrt{2}}
 \end{aligned}$$

$$\begin{aligned}
 \text{Area} &= \frac{1}{2} \times a \times a \\
 &= \frac{1}{2} \times \frac{h}{\sqrt{2}} \times \frac{h}{\sqrt{2}} \\
 &= \underline{\underline{\frac{h^2}{4}}}
 \end{aligned}$$



Hypotenuse = $2x = h$.

$$\begin{aligned}
 \therefore \text{Area} &= \frac{h^2}{4} = \frac{(2x)^2}{4} \\
 &= x^2.
 \end{aligned}$$

$$\begin{aligned}
 V &= \int_0^9 x^2 \, dy \\
 &= \int_0^9 (9 - y) \, dy \\
 &= \left[9y - \frac{y^2}{2} \right]_0^9 \\
 &= 81 - \frac{81}{2} \\
 &= \underline{\underline{\frac{81}{2} V^3}}
 \end{aligned}$$

$$\textcircled{7} \quad I_n = \int_0^{\pi/2} \cos^n x \, dx$$

$$= \int_0^{\pi/2} \cos x \cos^{n-1} x \, dx$$

Integration by Parts.

$$I_n = \left[\sin x \cos^{n-1} x \right]_0^{\pi/2} - \int_0^{\pi/2} \sin x (n-1) \cos^{n-2} x - \sin x$$

$$= + (n-1) \int_0^{\pi/2} \sin^2 x \cos^{n-2} x \, dx$$

$$= (n-1) \int_0^{\pi/2} (1 - \cos^2 x) \cos^{n-2} x \, dx$$

$$= (n-1) \left[\int_0^{\pi/2} \cos^{n-2} x \, dx - \int_0^{\pi/2} \cos^n x \, dx \right]$$

$I_{n-2} \qquad \qquad \qquad I_n$

$$I_n = (n-1) I_{n-2} + (n-1) I_n$$

$$n I_n = (n-1) I_{n-2}$$

$$\therefore I_n = \frac{n-1}{n} I_{n-2}$$

$$I_4 = \frac{3}{4} I_2$$

$$= \frac{3}{4} \times \frac{1}{2} I_0$$

$$= \frac{3}{8} \int_0^{\pi/2} \cos^0 x \, dx$$

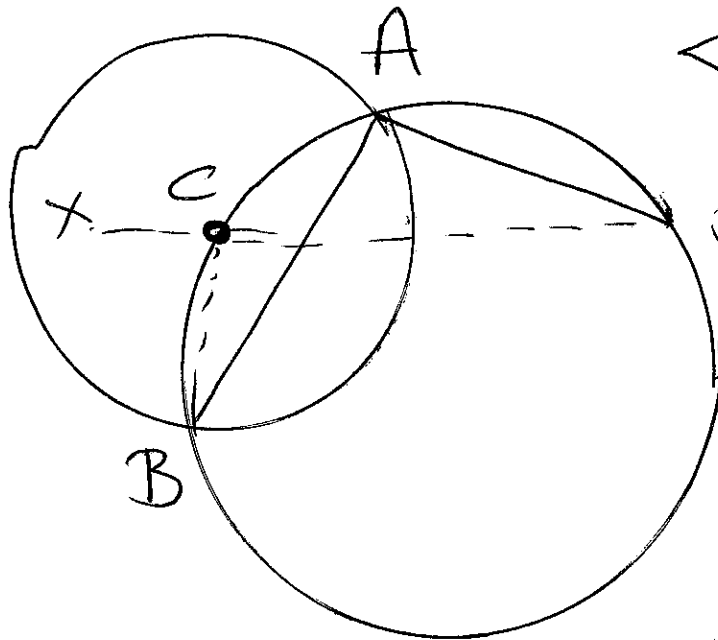
$$= \frac{3}{8} \int_0^{\pi/2} 1 \, dx$$

$$= \frac{3}{8} [x]_0^{\pi/2}$$

$$= \frac{3\pi}{16}$$

7b)

i)



$$\angle PAB = \theta$$

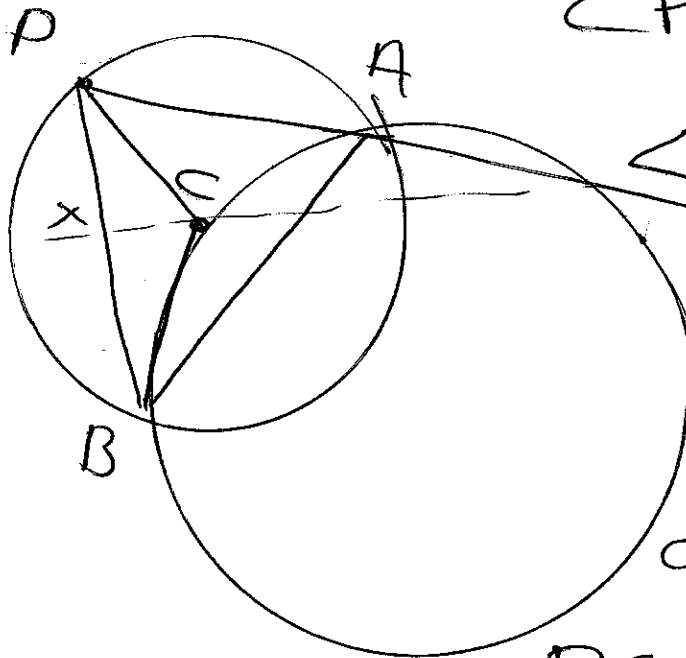
$$\angle PCB = \theta$$

(Angles on same arc)

$$\therefore \angle BCX = 180 - \theta$$

(Angles of a straight line)

ii)



$$\angle PAB = 180 - \theta$$

Angle of straight line

$$\angle PCB = 2\angle PAB$$

Angles at centre
= 2 × Angle at arc

$$\angle BCX = 180 - \theta$$

$$\therefore \angle PCX = 180 - \theta$$

$$\text{as } \angle PCB = 2(180 - \theta)$$

$$PC = BC \text{ Radii}$$

$$XC \text{ Common}$$

$$\therefore \triangle PCX \equiv \triangle BCX$$

$$\therefore PB \perp XC$$

$$(7) c) (a-b)^2 \geq 0$$

$$a^2 - 2ab + b^2 \geq 0$$

$$a^2 + b^2 \geq 2ab$$

$$(a+b)^2 - 2ab \geq 2ab$$

$$(a+b)^2 \geq 4ab$$

✓ as both +ve

$$a+b \geq 2\sqrt{ab}$$

$$\therefore \frac{a+b}{2} \geq \sqrt{ab}$$

$$a+b \geq 2\sqrt{ab} \quad a+c \geq 2\sqrt{ac} \quad b+c \geq 2\sqrt{bc}$$

$$\therefore (a+b)(a+c)(b+c) \geq 2\sqrt{ab} \cdot 2\sqrt{ac} \cdot 2\sqrt{bc} \\ \geq 8abc$$

$$\text{Let } A = b+c-a, B = a+c-b, C = a+b-c$$

$$(A+B)(A+C)(B+C) \geq 8ABC \quad \text{from above}$$

$$\text{but } AB = 2c \quad AC = 2b \quad BC = 2a$$

$$\therefore (b+c-a)(a+c-b)(a+b-c) = 8abc$$

$$\text{Cancel 8} \quad \therefore abc \geq (b+c-a)(a+c-b)(a+b-c)$$

$$\text{Given } [abc - a^2(b+c-a)] + [abc - b^2(a+c-b)] + [abc - c^2(a+b-c)] \\ \geq 0$$

$$\text{But } abc - a^2(b+c-a) = a(bc - ab - ac + a^2) \\ = a[bc - a + a(c-b)] \\ = a(a-c)(a-b)$$

LHS

$$a(a-b)(a-c) + b(b-a)(b-c) + c(c-a)(c-b)$$

$$(a-b)(a^2 - ac - b^2 + bc)$$

$$(a-b)(a^2 - b^2 - c(a-b))$$

$$(a-b)[(a-b)(a+b) - c(a-b)]$$

$$(a-b)^2(a+b-c) \geq 0$$

Q8₁ $\sum_{k=1}^n \frac{1}{(k+1)(k+2)} = \frac{n}{2(n+2)}$

SI $n=1 \quad \frac{1}{(2)(3)} = \frac{1}{2(3)} \quad \checkmark$

S2 Assume true $n=k \quad \sum_{k=1}^n \frac{1}{(k+1)(k+2)} = \frac{n}{2(n+2)}$

Show true $n=k+1$
 $\sum_{k=1}^{n+1} \frac{1}{(k+1)(k+2)} = \frac{n+1}{2((n+1)+2)} = \frac{n+1}{2(n+3)}$

$$\begin{aligned} \sum_{k=1}^{n+1} \frac{1}{(k+1)(k+2)} &= \sum_{k=1}^n \frac{1}{(k+1)(k+2)} + \frac{1}{((n+1)+1)((n+1)+2)} \\ &= \frac{n}{2(n+2)} + \frac{1}{(n+2)(n+3)} \\ &= \frac{n(n+3) + 2}{2(n+2)(n+3)} \\ &= \frac{n^2 + 3n + 2}{2(n+2)(n+3)} \\ &= \frac{\cancel{(n+2)}(n+1)}{2\cancel{(n+2)}(n+3)} \\ &= \frac{n+1}{2(n+3)} \end{aligned}$$

SB

Q. 8 ii) $\lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{1}{(k+1)(k+2)} = \lim_{n \rightarrow \infty} \frac{n}{2(n+1)}$
 from (i)

$$\lim_{n \rightarrow \infty} \frac{n}{n} \left(\frac{1}{2(1+k)} \right) = \frac{1}{2} \quad S = \frac{1}{2}$$

iii) $\int_1^{\infty} \frac{dx}{(x+1)(x+2)} = \int_1^{\infty} \frac{1}{(x+1)(x+2)} - \frac{1}{(x+2)}$
 (Partial Fraction)

$$= \left[\ln \frac{(x+1)}{(x+2)} \right]_1^{\infty}$$

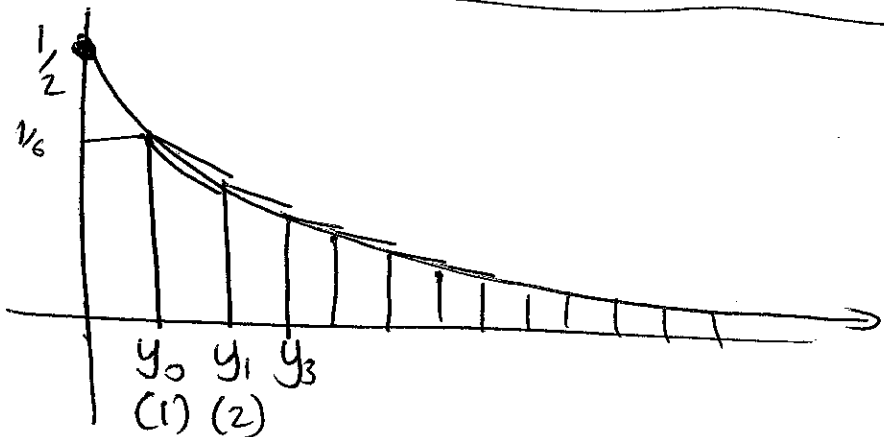
$$= \ln 1 - \ln \frac{2}{3}$$

and $x \rightarrow \infty$

$$= \ln \frac{3}{2}$$

$$I = \ln \frac{3}{2}$$

iv(v) $\frac{1}{(x+1)(x+2)}$



$$T = \lim_{n \rightarrow \infty} \frac{h}{2} (y_0 + y_n + 2(y_1 + y_2 + y_3 + \dots + y_{n-1}))$$

$$h=1 \quad y_0 = \frac{1}{6} \quad y_n = 0 \quad \left[\text{As } n \rightarrow \infty \right] \quad 2 \left(\sum_{k=2}^n \frac{1}{(k+1)(k+2)} \right)$$

$$T = \frac{1}{2} \left(\frac{1}{6} + 0 + \frac{1}{6} \right) \quad \frac{1}{2} - \frac{1}{6}$$

$$T = \frac{5}{12}$$

$$I < T < S$$